

NodeDesk VR: A Spatially Organized Focus View Navigation System for VR Desktop Environments

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Traditional desktop systems often flatten interconnected information into lists, overlapping windows, and tab changes, making repeated traversal between related concepts visually cluttered and cognitively fragmented. This paper presents NodeDesk VR, a virtual reality desktop environment for spatial knowledge organization and linked-node traversal. The prototype combines spatial node creation, direct node manipulation, dynamic node linking, a wrist-mounted control interface, and a Focus View mode for joystick-based navigation through connected nodes. A key design decision is that traversal direction is computed from the nodes' original graph-relative layout rather than their temporary presentation positions during Focus View. A formative qualitative study with six participants evaluated perceived intuitiveness, clutter reduction, spatial orientation, and repeated-traversal usability. The results suggest that spatial node arrangement and graph-relative joystick traversal were generally intuitive, while Focus View scale, abrupt transitions, headset fogging, and dense local graph structure produced the clearest usability limitations.

CCS Concepts: • **Human-centered computing** → **Virtual reality**; *Interaction design*; *Graphical user interfaces*; *Empirical studies in HCI*.

Additional Key Words and Phrases: virtual reality, VR desktop, spatial user interfaces, immersive analytics, knowledge graphs, graph traversal, focus view, XR productivity

Executive Summary

NodeDesk VR is a Unity-based VR desktop prototype for authoring, arranging, linking, and traversing information nodes in three-dimensional space. The system's central contribution is a Focus View interaction that improves local readability while preserving graph-relative directionality for joystick-based traversal. A six-participant formative evaluation found the strongest support for the intuitiveness of the spatial workspace and for directionally anchored traversal between linked nodes. Participants also identified several concrete problems: Focus View could feel too close or abrupt, the Meta Quest 2 headset imposed noticeable field-of-view and fogging constraints, and dense graphs made traversal less predictable. These findings position NodeDesk VR as a promising early exploration of spatially organized knowledge work in VR while also identifying specific design changes for future iterations.

1 Introduction

Desktop interfaces remain highly effective for conventional productivity tasks, but they typically represent interconnected information through tabs, lists, windows, and file hierarchies that obscure structural relationships. When users repeatedly move among related documents, notes, or concepts, the workflow can become fragmented because links are implicit rather than spatially visible.

Virtual reality (VR) offers a different design space. Information can be arranged spatially, revisited through embodied action, and explored using persistent three-dimensional context. Prior work on spatial document management and immersive workspaces suggests that layout, embodiment, display scale, and body-centric interaction can affect how users organize, recall, and navigate information [1, 3, 4, 10, 12, 13, 15, 17–19, 21].

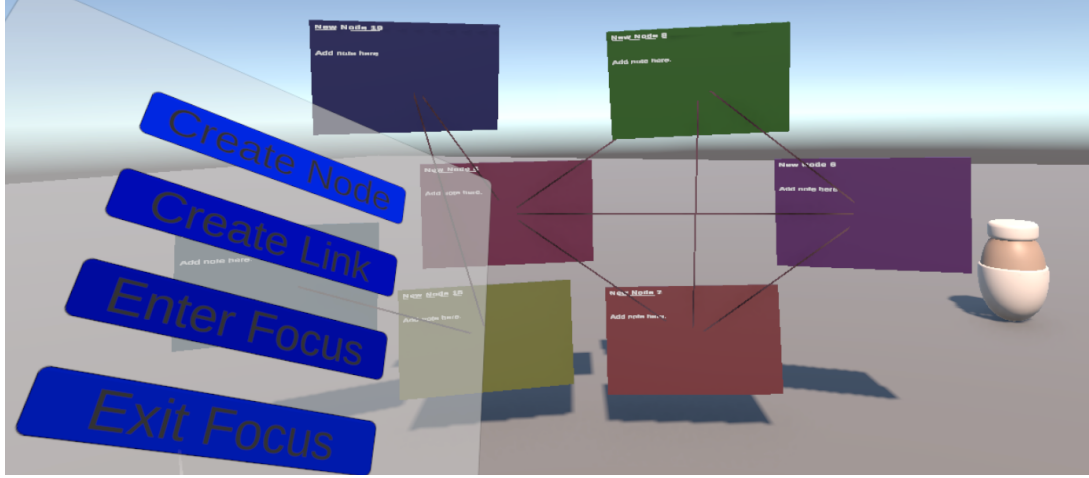


Fig. 1. Main NodeDesk VR workspace showing spatially arranged nodes, visible links, and the wrist-mounted menu.

This paper presents *NodeDesk VR*, a prototype VR desktop for organizing and traversing interconnected nodes. Users can create node panels, position them in space, connect them with explicit links, and enter a Focus View mode that enlarges the current node for readability while preserving meaningful spatial relationships to its neighbors.

The work investigates the following research hypothesis:

A spatially organized Focus View navigation system in a VR desktop environment may improve navigation efficiency during repeated traversal of interconnected information networks.

The paper contributes three main elements. First, it describes a VR desktop interface that combines spatial node organization with graph-based traversal. Second, it introduces a Focus View interaction in which directional traversal remains anchored to the authored graph layout rather than to temporary presentation transforms. Third, it reports a formative evaluation that identifies both the promise and the current usability limits of spatially organized repeated traversal in immersive desktop environments.

2 Related Work

2.1 Spatial Document and Window Management

Spatial organization has long been used to support memory and retrieval in interface design. Data Mountain demonstrated that arranging documents within a spatial 3D environment can improve document management by leveraging users' spatial memory [17]. The Task Gallery extended these ideas into task and window management by using a 3D gallery metaphor for task switching and comparison [18]. On head-worn displays, the Personal Cockpit further showed that placing task-relevant windows around the user's body can support task switching while taking advantage of spatially distributed content [4].

Contemporary XR systems have revisited these questions using modern head-mounted displays and more ergonomic interaction techniques. Projective Windows introduced a way to grab, move, and position windows in space using direct spatial interaction [10]. Expanding the Bounds of Seated Virtual Workspaces studied how large virtual work surfaces can support knowledge work while improving comfort and layout flexibility [13]. Breaking the Screen explored how

touchscreen boundaries can be extended into VR for mobile knowledge work [1]. More recent systems have studied situated workspace adaptation, high-resolution VR home-office work, ergonomic layout support, spatial arrangement of augmented-reality documents, virtual display strategies, palm-based window management, and XR window managers for 2D applications in immersive space [3, 11, 12, 14, 15, 19, 21].

NodeDesk VR builds on this literature but shifts the emphasis from generic application windows toward authored note-like nodes and explicit semantic links between them. The design focus is not only placement, but repeated directional traversal among retained spatial relationships.

2.2 Immersive Analytics and Embodied Navigation

Immersive analytics frames VR and AR as environments for analytical reasoning and visual exploration rather than only simulation or entertainment [20]. Within this area, prior work has examined how overview-plus-detail strategies and embodied navigation affect understanding of large visual information spaces. Yang et al. compared overview+detail and zooming approaches for immersive 3D scatterplots [22]. Huang et al. studied the benefits of embodied interaction when navigating network visualizations, reporting nuanced trade-offs among performance, preference, and spatial understanding [5].

These studies are relevant because NodeDesk VR also balances overview and detail. However, instead of navigating a single large visualization by moving through space, NodeDesk VR emphasizes authored linked information panels and repeated return paths through a spatial information graph.

2.3 Graph Interaction and Knowledge-Graph Exploration in VR

Recent work has focused more directly on graph and network interaction in immersive environments. Visual Comparison of Networks in VR investigated immersive techniques for comparing network structures [8]. Aesthetic-Driven Navigation for Node-Link Diagrams in VR introduced viewpoint guidance strategies for better graph reading [7]. Evaluating Node Selection Techniques for Network Visualizations in Virtual Reality compared six node selection methods and showed how graph density and clutter can strongly influence usability [6].

Beyond generic node-link diagrams, immersive systems have also been proposed for linked-data and RDF graph exploration. Kellmann et al. presented Graph2VR for exploring linked data in VR [9], while Quantin et al. introduced AIDEN for immersive exploration and manipulation of RDF knowledge graphs [16]. Most recently, Büschel et al. studied how spatial context switches affect document-centric knowledge tasks in extended reality, reinforcing the importance of layout continuity in XR knowledge work [2].

NodeDesk VR differs from these systems by explicitly combining a VR desktop metaphor, user-authored spatial organization, explicit linking, and a Focus View traversal technique whose directional semantics remain tied to the original graph layout even when the focused node is moved for readability.

3 System Design

NodeDesk VR was designed around four goals:

- **Preserve spatial memory.** Users should be able to distribute information in a way that remains meaningful over time.
- **Reduce local clutter.** Dense spatial layouts should still support readable local inspection.

Table 1. Core interaction components in NodeDesk VR.

Component	Description
Spatial node creation	Instantiates world-space node panels from the wrist-mounted menu.
Spatial node arrangement	Allows users to move nodes into semantically meaningful spatial groupings.
Dynamic linking	Connects two nodes with a persistent line-rendered edge.
Focus View	Enlarges and centers the currently selected node for local readability.
Directional traversal	Uses joystick input and original graph-relative node offsets to select the next linked node.
Trash/delete interaction	Removes nodes from the workspace through a world-space delete affordance during free interaction.

- **Support repeated traversal.** Users should be able to revisit related nodes multiple times without losing directional understanding.
- **Remain lightweight on standalone hardware.** The interaction model should function using standard Meta Quest controller input without external tracking infrastructure.

3.1 Spatial Node Workspace

NodeDesk VR represents information as floating node panels placed in a room-scale virtual workspace. Each node stores a title, body text, color, and a set of link relationships. Rather than enforcing a single layout structure, the system allows users to arrange nodes according to their own semantic logic—for example by topic, hierarchy, or sequence.

This spatial arrangement is not merely cosmetic. The authored layout becomes the underlying reference frame used later during Focus View traversal. Consequently, the workspace functions simultaneously as a visual organization surface and as a traversal map.

3.2 Dynamic Linking

Users can define explicit relationships between nodes through a link-creation mode. Once linked, nodes are connected through a persistent visual edge rendered in world space. These links transform the workspace from a collection of floating notes into a spatial graph. In the current prototype, links primarily indicate navigable relationships rather than typed semantic relations, but the structure is compatible with future extensions such as typed or directional links.

3.3 Wrist-Mounted Controls

The primary system controls are attached to the non-dominant wrist as a compact world-space menu. This menu supports actions such as creating nodes, toggling link mode, entering Focus View, and performing basic state transitions. A wrist-mounted menu was chosen to keep controls close to the body, avoid a permanently screen-centered HUD, and maintain compatibility with direct manipulation in the surrounding space.

3.4 Focus View

Focus View is the main navigation mode investigated in this project. When activated on a node, the selected node is enlarged and repositioned in front of the user for readability. This local presentation is intended to reduce surrounding clutter and make one node the temporary center of interaction.

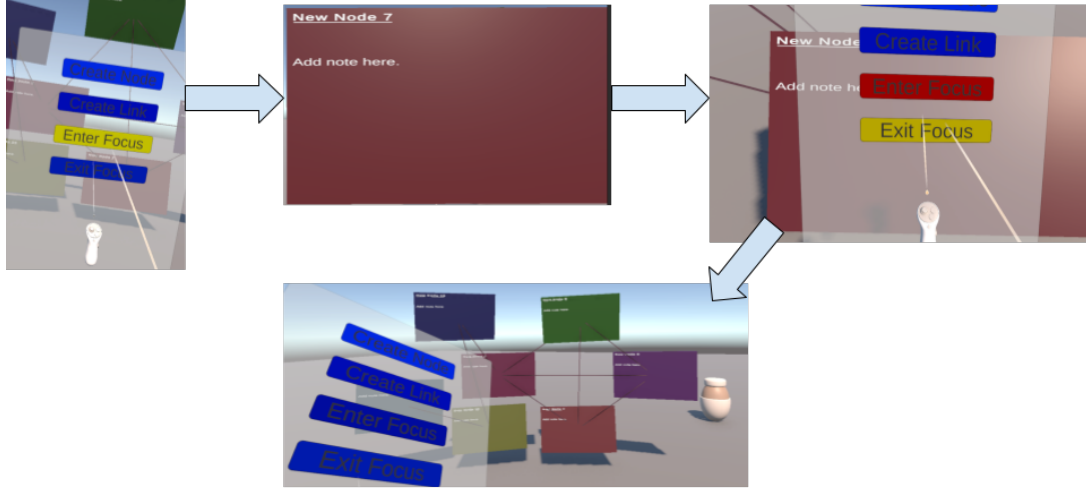


Fig. 2. Focus View sequence showing a selected node enlarged for local readability while preserving graph-relative relationships.

The design intentionally separates *presentation position* from *graph position*. The node shown in front of the user is not treated as though the graph itself has been rearranged. Instead, the system temporarily presents the node in a more readable location while retaining the original spatial layout as the basis for subsequent traversal.

3.5 Spatially Directed Traversal

While in Focus View, users traverse linked neighbors using joystick input. The central design decision is that joystick direction is interpreted against the node graph’s original spatial arrangement rather than the node’s temporary front-of-user presentation transform.

When traversal input occurs, the system evaluates the set of linked neighbors for the currently focused node and compares the user’s joystick direction to each neighbor’s stored graph-relative direction. The neighbor whose graph-relative vector most closely matches the joystick input becomes the next focused node. This allows repeated traversal to remain spatially consistent: a node authored to the left of the current node continues to behave like a leftward traversal target even after the current node has been moved in front of the user for readability.

4 Implementation

NodeDesk VR was implemented in Unity using C# and the XR Interaction Toolkit for Meta Quest 2 hardware. The prototype uses world-space user interface panels for nodes, direct controller-based manipulation, and joystick input for traversal in Focus View.

4.1 Software Architecture

At the software level, the project is organized around several core scripts. `NodeDeskManager` coordinates workspace-level actions such as node creation and system state changes. `NodePanel` stores per-node text, colors, and display behavior. `NodeSelectable` manages controller selection and focus eligibility. `LinkManager` and `NodeLink` maintain

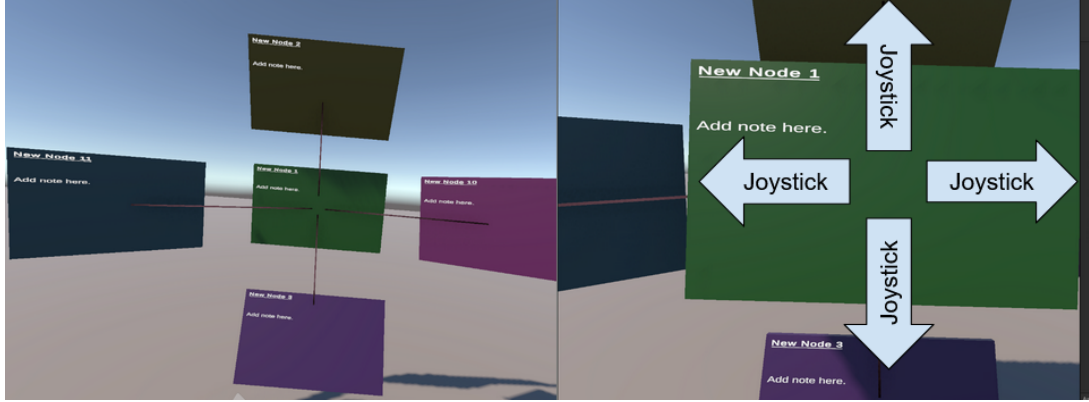


Fig. 3. Joystick-based directional traversal between linked nodes using the original spatial graph layout.

visual and logical relationships between nodes using persistent world-space line renderers. FocusViewManager handles Focus View entry and exit, restoration of original transforms, and directional traversal across linked neighbors.

A key implementation detail is that Focus View caches the original transforms of nodes before any temporary repositioning occurs. Traversal logic then references these retained graph-relative relationships rather than the temporary presentation pose. This distinction is central to the prototype’s research contribution.

4.2 Interaction Features

The current prototype supports the full interaction cycle used in the study: node creation from the wrist menu, direct movement of nodes in world space, explicit link creation, Focus View entry and exit, joystick-based graph traversal, and node deletion through a trash affordance. Nodes are currently optimized for short note-like content rather than full document editing. The prototype therefore prioritizes spatial organization and traversal over rich text authoring or large-scale project management.

4.3 Reproducibility

For reproducibility, the final project package should include the Unity source project, the final report PDF, the full $\text{\LaTeX}$ source, referenced literature PDFs, a README describing controller mappings and build steps, and media illustrating the prototype. The architecture, scripts, and study assets described here are sufficient to reproduce the study scenario, subject to inclusion of the final repository URL and project build instructions.

5 Evaluation Design

5.1 Research Questions

The formative study was organized around three research questions:

- **RQ1:** Does spatial organization help users understand relationships between information nodes?
- **RQ2:** Does Focus View reduce clutter and improve local readability during graph traversal?
- **RQ3:** Is joystick-based directional traversal intuitive for repeated navigation between linked nodes in VR?

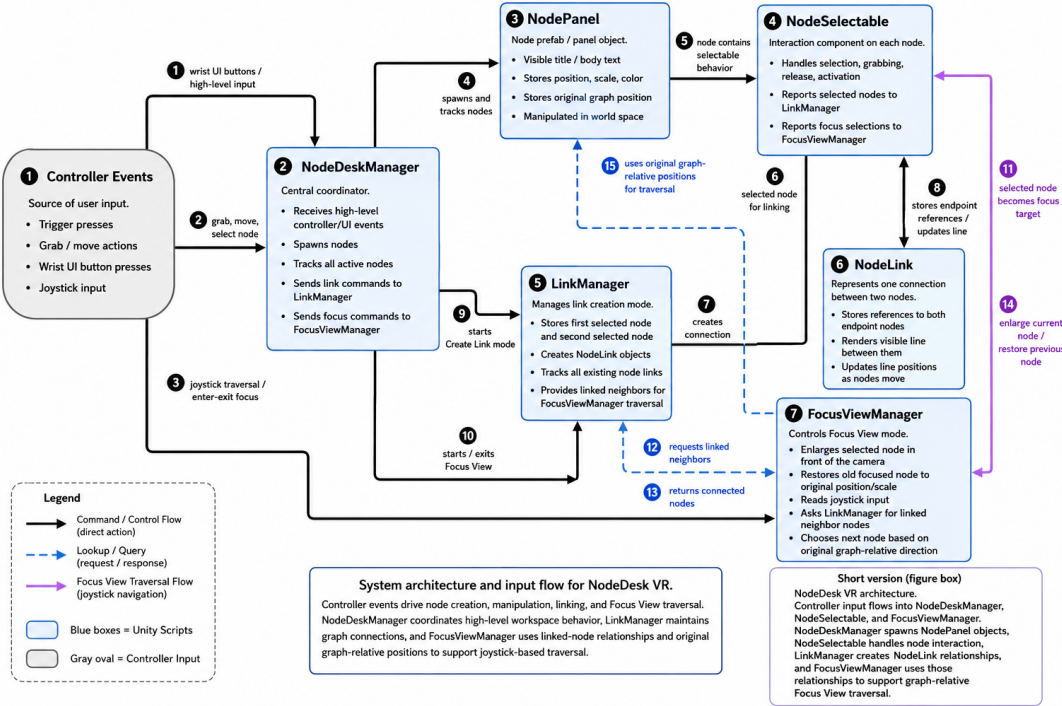


Fig. 4. NodeDesk VR system architecture showing relationships among NodeDeskManager, NodePanel, NodeSelectable, LinkManager, NodeLink, and FocusViewManager.

5.2 Study Design

The evaluation was designed as a formative qualitative study supported by post-study Likert responses and observational notes. Participants completed a common task sequence, verbalized impressions during use when possible, and then answered a short questionnaire plus open-ended reflection prompts. This structure was chosen because the prototype was still in an exploratory stage and the immediate goal was to identify usability breakdowns, promising interaction patterns, and design implications for future iterations rather than to produce confirmatory benchmark results.

6 Methods

6.1 Participants

Six participants completed the study over two days in May 2026. Individual sessions lasted approximately 13–15 minutes based on the recorded session times available for five of the six participants. Exact demographic fields such as age, gender, and prior VR experience distribution should be inserted from the finalized participant records. One participant (P6) explicitly exhibited a stronger VR learning curve than the rest of the sample, which meaningfully influenced that participant's questionnaire ratings and observational outcomes.

6.2 Apparatus

The study used the NodeDesk VR prototype running on a Meta Quest 2 headset with standard motion controllers. Participants interacted in a clear indoor test area and used the wrist-mounted menu, grab interactions, trigger input, and joystick traversal configured in the Unity build. During the sessions, several participants noted hardware-related limitations of the headset itself, especially limited field of view and lens fogging.

6.3 Tasks and Procedure

Each session followed the same general sequence:

- (1) Receive a brief introduction to the workspace and controls.
- (2) Create several nodes using the wrist-mounted interface.
- (3) Arrange the nodes spatially in ways that felt meaningful.
- (4) Create links between nodes judged to be related.
- (5) Enter Focus View on one node.
- (6) Use joystick traversal to navigate between linked nodes.
- (7) Exit Focus View and return to the authored workspace.
- (8) Engage in brief exploratory interaction and complete the post-study questionnaire.

The researcher recorded observational notes during each session and documented short participant responses about what felt intuitive, what felt confusing, and what changes participants would recommend.

6.4 Measures

The study combined three forms of evidence:

- direct observation of task performance and breakdowns,
- a seven-item Likert questionnaire using a 1–5 scale, and
- short open-ended comments gathered at the end of the session.

The 1–5 questionnaire used the scale anchors *1 = Strongly Disagree* and *5 = Strongly Agree*. Observations were used to contextualize the numeric responses, especially in cases where headset comfort, blurriness, or unfamiliarity with VR affected performance.

6.5 Questionnaire Items

The exact post-study protocol questions were:

- (1) The VR workspace felt intuitive to use.
- (2) The spatial arrangement of nodes helped me understand relationships between information.
- (3) Focus View reduced clutter and made information easier to navigate.
- (4) Joystick-based navigation between linked nodes felt natural.
- (5) I could maintain my sense of orientation while navigating between nodes.
- (6) The interaction techniques were easy to learn.
- (7) The system encouraged me to think spatially about connected information.

6.6 Data Analysis

Likert responses were summarized descriptively because the study was small and formative rather than confirmatory. Observation notes and short participant comments were analyzed thematically around four recurring topics: workspace intuitiveness, readability and clutter, orientation and transition behavior, and graph scalability. No inferential statistics were conducted for this initial sample.

7 Participant Protocol

7.1 Session Flow

The protocol was intentionally lightweight so that participants could experience the full interaction cycle of NodeDesk VR in a short session. Participants began by creating and moving nodes, proceeded to graph linking, then used Focus View and joystick traversal, and finished with free interaction plus short reflective feedback. This structure ensured that every participant encountered the same core system components while still allowing space for spontaneous exploratory behavior.

7.2 Task Checklist

The participant-facing task checklist can be summarized as follows:

- (1) Use the wrist menu to create at least two nodes.
- (2) Move the nodes into a spatial organization that makes sense to you.
- (3) Create links between related nodes.
- (4) Select a node and enter Focus View.
- (5) Use the joystick to move between linked nodes.
- (6) Exit Focus View and return to the workspace.
- (7) Explore any interaction that seems useful or interesting.

7.3 Post-Task Reflection

After the interactive portion, participants completed the seven Likert items listed in Section 8.3 and answered short reflective prompts about the most intuitive interaction, the most confusing interaction, the usefulness of Focus View, and design improvements they would recommend. These open-ended comments proved especially useful for identifying transition discomfort, headset-related visibility challenges, and graph-density limits.

8 Results

8.1 Participant Overview

The six study sessions collectively suggest that participants could understand the overall concept of NodeDesk VR after a short introduction, but the quality of that understanding depended strongly on both VR familiarity and the complexity of the authored graph. Five of the six participants adapted quickly to the workspace and reported positive impressions of spatial arrangement, whereas one participant (P6) struggled primarily with the VR controls and interface itself rather than the larger concept of spatial organization. Across participants, the clearest positive reactions concerned node movement and directional traversal. The clearest negative reactions concerned Focus View scale, abrupt transitions, optical discomfort, and dense local graph structure.



Fig. 5. Study setup using the Meta Quest 2 headset and motion controllers during participant evaluation.

Table 2. Qualitative participant summary from observational notes and short reflective responses.

Participant Session	Node creation / spatial arrangement	Focus View / clutter	Navigation / orientation	Open-ended comments and suggestions
P1 3 May 2026, 2:35–2:48 PM	Understood creation and movement quickly; described moving nodes as “fun and pretty intuitive.”	Understood Focus View and said it improved readability, but also reported that it was “too close and too big.”	Understood joystick traversal and expected it to match the authored layout; did not remain oriented after exiting Focus View.	Most intuitive interaction was moving nodes; most confusing interaction was linking.
P2 3 May 2026, 2:50–3:05 PM	Understood node creation and arranged nodes by color.	Understood the transition and reported improved readability.	Remained oriented after exiting Focus View.	Reported headset fogging (“Steam makes it blurry”); suggested adjustable Focus View distance and the ability to rotate nodes.
P3 3 May 2026, 3:30–3:44 PM	Noted that nodes sometimes spawned on top of one another.	Understood Focus View but did not think it improved readability because of blurriness.	Reported joystick traversal was faster than moving among physical notes.	Liked the trash can interaction; suggested extending the system with apps rather than only notes.
P4 3 May 2026, time not recorded	General workflow was understood, but the main reaction focused on the view transition.	Described Focus View as jarring (“Suddenly in my face”) and felt it blocked the view of everything else.	Described traversal as easy “warping” between nodes, but gave a lower orientation rating after traversal.	Suggested smoother animated transitions to preserve orientation.
P5 May 2026, 6:05–6:20 PM	Successfully created a complex web of linked nodes.	Understood Focus View and rated clutter reduction positively.	Remained oriented after exiting, but reported traversal became unpredictable when the graph became too dense.	Recommended limiting each node to roughly four links to maintain navigability.
P6 4 May 2026, 1:30–1:44 PM	Only somewhat understood node creation and linking because of the VR learning curve.	Understood Focus View after discovering the lower trigger input; still reported interface confusion more broadly.	Only somewhat remained oriented after exiting Focus View; low traversal rating reflected difficulty with VR controls.	Said the VR buttons and wrist UI were confusing; suggested making the system an AI-assisted environment.

8.2 Observed Interaction Patterns

Direct manipulation of nodes was the most consistently successful part of the prototype. P1 explicitly described node movement as “fun and pretty intuitive,” and P2 immediately adopted a personal organizational strategy by arranging nodes according to color. Even P5, who later encountered complexity problems, was able to author a large interconnected graph. These observations suggest that the workspace successfully supported the *creation* of spatial structure.

Link creation was less immediately obvious. P1 initially reported, “It is not working,” before understanding the interaction sequence, and P6 similarly struggled with the general VR control scheme. These observations suggest that link creation, not spatial arrangement itself, was the more fragile learnability point early in the session.

Focus View produced a more mixed pattern. Some participants valued the enlarged presentation for readability and clutter reduction, while others felt that the transition was visually abrupt or optically uncomfortable. P1 said the

Table 3. Post-study Likert responses using the exact questionnaire items from the participant protocol. Scale: 1 = Strongly Disagree, 5 = Strongly Agree.

Question	P1	P2	P3	P4	P5	P6	Mean
The VR workspace felt intuitive to use.	4	5	5	5	5	1	4.17
The spatial arrangement of nodes helped me understand relationships between information.	5	4	5	4	5	2	4.17
Focus View reduced clutter and made information easier to navigate.	5	3	3	3	5	5	4.00
Joystick-based navigation between linked nodes felt natural.	5	5	5	5	5	2	4.50
I could maintain my sense of orientation while navigating between nodes.	4	5	5	3	5	2	4.00
The interaction techniques were easy to learn.	5	5	5	4	5	1	4.17
The system encouraged me to think spatially about connected information.	5	4	5	4	5	2	4.17

focused node was “too close and too big,” and P4 described the experience as “suddenly in my face.” P2 and P5 reacted more positively, but P2’s comments were confounded by headset fogging and reduced clarity.

Joystick traversal produced some of the clearest positive signals in the study. P3 said traversal was faster than moving among physical notes, and P4 described it as an efficient way to “warp” between nodes. These comments suggest that, once participants internalized the mapping between joystick direction and the authored layout, the traversal technique conveyed a strong sense of intentional movement through related information.

8.3 Post-Study Likert Responses

Table 3 reports the seven questionnaire items using the exact protocol wording. Across all 42 item responses, the overall average was 4.17/5. The highest-rated prompt was *Joystick-based navigation between linked nodes felt natural* with a mean of 4.50. Four additional items—workspace intuition, spatial arrangement, ease of learning, and encouragement of spatial thinking—each averaged 4.17. The two most mixed items were clutter reduction in Focus View and maintenance of orientation, each averaging 4.00.

Five of the six participants rated the workspace as intuitive (4 or 5), and five of the six also rated the spatial arrangement question positively (4 or 5). This pattern is consistent with the observational notes, which show that most participants quickly appropriated the workspace as a spatial map of relationships.

Focus View responses were more divided: three participants gave the clutter-reduction item a 5, while three gave it a 3. This reflects the tension visible in the comments. Conceptually, users tended to understand the value of enlarging the active node, but some experienced the implementation as too abrupt, too close, or optically degraded by the headset.

Joystick traversal was the strongest numeric result. Five participants rated it a 5, while only P6 gave it a low rating. This result is notable because it suggests that the graph-relative directional logic aligned well with user expectations once the broader interaction model was understood. Orientation responses were positive overall but less uniform, with two lower scores reflecting exactly the disruption that participants described in relation to the Focus View transition.

8.4 Emergent Themes

Four themes emerged from the combined observations and questionnaire results.

Spatial arrangement was immediately meaningful. Participants generally treated the workspace as a conceptual map rather than as a pile of disconnected notes. P2 grouped content by color, P5 built a “complex web,” and the spatial arrangement item averaged 4.17. These observations suggest that NodeDesk VR successfully externalized relationships into space, consistent with prior work on spatial memory and spatial productivity interfaces [4, 13, 17].

Focus View improved local readability for some users but weakened continuity for others. Several participants appreciated the larger focused node, but the transition could feel abrupt or occluding. Participant comments suggest that the issue was not the concept of Focus View itself, but the current transition design and scale parameters. This is reinforced by the split responses to the clutter-reduction question.

Graph-relative joystick traversal was intuitive in small-to-moderate graphs. Participants generally understood that traversal direction should correspond to authored layout. The strongest ratings in the study supported this interpretation. However, P5’s dense graph revealed that the technique becomes less predictable when too many neighbors compete within similar directional sectors, a pattern consistent with prior findings about density and clutter in immersive node interaction [6].

Hardware limitations mattered. Several participants noted limitations caused by the Meta Quest 2 rather than purely by software design. The most common problems were restricted field of view, fogging, and blur. P2 explicitly described the headset as becoming steamy, while P3 associated readability problems with blurriness. These observations indicate that some of the negative reactions to Focus View were likely amplified by hardware constraints.

8.5 Placeholder for Expanded Quantitative Analysis

A larger follow-up study could add nonparametric analyses, confidence intervals, subgroup comparisons by VR familiarity, and controlled comparisons against non-spatial traversal baselines.

9 Discussion

The study provides early evidence that a spatially organized VR desktop can make linked information feel navigable as a structured environment rather than as a flat collection of disconnected notes. Participants were especially positive about node movement, spatial arrangement, and directionally anchored traversal. Together, these results suggest that the central design idea of preserving graph-relative direction during Focus View was understandable and often useful.

With respect to **RQ1**, the findings support the claim that spatial organization helped participants understand relationships between nodes. Five of six participants gave positive ratings to the spatial arrangement item, and the qualitative data showed natural grouping strategies such as organizing by color or constructing a linked web. This aligns with prior work showing that spatial arrangement can support memory, sensemaking, and task switching [2, 4, 13, 17].

With respect to **RQ2**, the findings were mixed. Focus View did appear to reduce local clutter for some participants, but not reliably for all of them. The mixed responses were tied less to the abstract idea of focusing one node and more to the concrete implementation details: Focus View could appear too close, too sudden, or visually unclear because of the headset. This suggests that local readability in immersive desktop systems depends not only on spatial separation, but also on transition design, ergonomic calibration, and headset optics [14, 15].

With respect to **RQ3**, the results were most clearly positive. Joystick traversal received the strongest ratings in the study, and several participants explicitly described it as fast or natural. The technique therefore appears promising for repeated traversal through small linked neighborhoods in a spatial information graph. At the same time, P5’s complex

graph demonstrates that authored density can degrade the clarity of directional traversal. This indicates that future work should distinguish between *local traversal usability* and *global graph scalability*.

An important interpretive point is that several negative reactions were not primarily conceptual failures. They were implementation or device failures: abrupt transitions, node scale calibration, fogging, blur, and the limited field of view of the Quest 2. This matters because it suggests the design hypothesis remains viable even though the present prototype needs ergonomic refinement.

10 Limitations

This work has several limitations. First, the study used a small six-participant formative sample, so the numeric results should be interpreted descriptively rather than inferentially. Second, the prototype focused on note-like spatial nodes and traversal logic rather than full productivity features such as rich text editing, search, metadata, or document import. Third, the evaluation was conducted on Meta Quest 2 hardware, and multiple usability issues were influenced by the headset’s optical and comfort limitations.

A further limitation is that the system currently treats links as visually equivalent and largely unrestricted. As P5’s session showed, this can produce ambiguous traversal targets when multiple neighbors occupy similar directional positions. The current results therefore characterize an early but incomplete design space.

11 Future Work

Several concrete next steps follow directly from the results. The most immediate improvement is to add smoother transition animations when entering Focus View and when traversing from one focused node to the next. Participants repeatedly indicated that abrupt visual changes weakened orientation. Adjustable Focus View distance and scale should also be added so that users can tune readability without pushing the focused node uncomfortably close.

Graph scalability should be addressed more directly. One practical change would be to constrain the number of links per node—for example, to four neighbors—so that each direction remains distinct. More sophisticated strategies could include link previews, semantic clustering, typed links, mini-maps, or explicit neighborhood disambiguation when multiple targets compete for the same direction.

Future versions of NodeDesk VR should also expand beyond note panels. Participants suggested richer applications, rotatable objects, and AI-assisted functionality. These suggestions point toward a broader immersive desktop environment in which authored nodes coexist with application windows, structured knowledge artifacts, or AI-generated summaries. Comparative studies against menu-based traversal, gaze-based selection, or traditional desktop navigation would also help isolate the specific contribution of graph-relative directional traversal.

12 Conclusion

This paper presented NodeDesk VR, a spatially organized VR desktop prototype for creating, linking, and traversing information nodes in three-dimensional space. The system centers on a Focus View navigation technique that enlarges the current node for local readability while preserving the original graph-relative layout as the basis for directional traversal.

A six-participant formative evaluation found strong support for spatial arrangement and joystick-based traversal, suggesting that users can understand and benefit from directionally anchored navigation in an immersive node workspace. At the same time, the study surfaced clear design limits related to transition abruptness, headset visibility

constraints, and the reduced navigability of dense graphs. Overall, the findings indicate that spatially organized Focus View navigation is a promising direction for VR desktop systems aimed at linked information work.

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